

Methods of Enumeration

ECON 97: Economic Toolkit

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<https://join.iclicker.com/RDNG>

Quiz: Sets

Suppose that A and B are two events with $P(A) = 0.4$ and $P(B) = 0.5$. Which of the following can NOT be a possible value of $P(A \cup B)$?

A. 0.4

B. 0.5

C. 0.6

D. 0.7

E. 0.9

No in-person class Thursday, April 9 & no office hours April 10

→ R coding tutorial will be posted instead, based on the prompt below

Optional coding challenge:

Write an R script that simulates drawing a hand of 5 cards from a standard 52-card deck. Try to find a way to simulate many draws and estimate the probability of various poker hands (e.g., full house, straight, pair)

Skills to practice:

- Generate random samples
- Define objects
- Dynamically store results
- Compute summary statistics

Review of Concepts

Important formulas:

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- Compliment: $P(A') = 1 - P(A)$
- Mutually exclusive: $P(A \cap B) = 0$
- Exhaustive: $P(A \cup B) = 1$
- $P(A \cup B)' = P(A' \cap B')$ and $P(A \cap B)' = P(A' \cup B')$
- $P(A \cap B') = P(A) - P(A \cap B)$
- Subset: if $A \subset B$ then $P(A \cap B) = P(A)$

Methods of Enumeration

How to count the number of possible outcomes of an experiment?

E.g.: flip a coin 3 times :

$$\{HHH, HHT, HTH, HTT, THH, TTH, THT, TTT\}$$

8 possible combinations of Heads and Tails.

It's not always feasible to list every outcome. We need a formal mathematical approach to count them.

In general, look at each experiment as a series of stages; at each stage, there is a certain number of possible outcomes.

E.g., for 3 coin flips, each flip is a stage with 2 possibilities (H or T)

$$2 * 2 * 2 = 8$$

How to count

Two important questions:

① Is there replacement?

- Can outcomes repeat at each stage?
- E.g., coin tosses have replacement, because H and T can repeat
- E.g., drawing cards from a deck has no replacement if you don't put drawn cards back

② Does the order matter?

- Do the outcomes change if you re-order the possibilities?
- E.g., security PIN code: 1457 and 1475 are distinct codes
- E.g., number of ingredient combinations for a recipe, order doesn't matter

How to count

Define the following:

n : number of options available

k : number of selections made (sample size)

1 Replacement

- Order matters: n^k
- Order doesn't matter: not covered in Econ 41

2 No replacement

- Order matters: ${}_nP_k = \frac{n!}{(n-k)!}$
- Order doesn't matter: ${}_nC_k = \frac{n!}{(n-k)!k!}$

Practice example

How many different ways can we arrange the letters in “MISSISSIPPI”?

Practice example

Suppose we roll a fair 6-sided die 6 times. What are the following probabilities?

- All possible numbers (1, 2, 3, 4, 5, 6)?
- Three 6s and three 1s?
- Two 1s, two 3s, and two 5s?
- Six even numbers?